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CONTAINER CAP

Abstract

A vented container cap configured to be operably coupled to a container such as but not limited to an intermediate bulk container wherein the present invention provides venting of the container and further inhibits coagulation of product so as to avoid clogging of vents. The invention includes a body that has an upper portion integrally formed with a lower portion wherein the lower portion is perpendicular to the upper portion. The lower portion includes a wall that is cylindrical in shape having a bottom member distal to the upper portion. The bottom member includes a lower aperture and void formed therein. An annular passage is formed in the wall member and is fluidly coupled to the void and lower aperture. Four vent apertures are fluidly coupled to the annular passage and are located at the upper end of the lower portion adjacent the lower surface of the upper portion.

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Background/Summary

FIELD OF THE INVENTION

[0001] The present invention relates generally to container caps, more specifically but not by way of limitation, a cap for a rigid intermediate bulk container wherein the cap of the present invention provides venting of the interior volume of the intermediate bulk container and further inhibits coagulation of the product in the vent cap of the present invention.

BACKGROUND

[0002] Intermediate bulk containers also known as IBC totes are industrial grade containers engineered for the mass handling, transport, and storage of liquids, semi-solids, pastes, or other materials. The two main categories of IBC totes are flexible IBC's and rigid IBC's. Rigid intermediate bulk containers are stackable, reusable, versatile containers with an integrated pallet base mount that provides maneuverability of the IBC tote with a pallet jack or forklift. The IBC totes can be made from metal, plastic, or a composite construction of the two materials. Rigid IBC design types are manufactured across a volume range that is in between that of standard shipping drums and intermodal tank containers. IBC totes are authorized to be fabricated of a volume up to 3 cubic meters (793 US gallons) and still be considered an IBC tote for shipping and handling permits.

[0003] Intermediate bulk containers are standardized shipping containers often Department of Transportation (DOT) certified for the transport handling of hazardous and non-hazardous materials. Some exemplary materials for which IBC totes are employed to store and ship are bulk chemicals, food syrups, solvents and liquid fertilizers. All rigid IBC totes are equipped with a cap on the top that provide access to fill and these caps are further configured with a conventional vent as some of the products require venting subsequent being stored in the IBC tote. Products such as but not limited to liquid fertilizers can coagulate when vented which can lead to clogged vent caps creating a potentially hazardous issue. Conventional vent caps are not configured to permit venting of an IBC tote and inhibit coagulation of a product in the vented cap.

[0004] Accordingly, there is a need for a vented cap for an IBC tote wherein the cap of the present invention permits venting of the interior volume of the IBC and further inhibits coagulation of the materials in the passages of the vented cap.

SUMMARY OF THE INVENTION

[0005] It is the object of the present invention to provide a vented cap for use on an intermediate bulk container wherein the cap of the present invention includes a body manufactured from plastic or other suitable material.

[0006] Another object of the present invention is to provide a cap for an intermediate bulk container that provides venting of the interior volume and further inhibits coagulation of product within the cap wherein the body of the present invention includes an upper portion and a lower portion.

[0007] A further object of the present invention is to provide a vented cap for use on an intermediate bulk container wherein the upper portion is perpendicular to the lower portion and integrally formed therewith.

[0008] Yet a further object of the present invention is to provide a cap for an intermediate bulk container that provides venting of the interior volume and further inhibits coagulation of product within the cap wherein the lower portion is cylindrical in shape having a lower surface distal to the upper portion.

[0009] Still another object of the present invention is to provide a vented cap for use on an intermediate bulk container wherein the lower portion includes an aperture formed therein proximate the lower surface.

[0010] An additional object of the present invention is to provide a cap for an intermediate bulk container that provides venting of the interior volume and further inhibits coagulation of product within the cap wherein the lower portion includes an annular passage in the wall of the lower

portion wherein the annular passage is fluidly coupled to the aperture.
[0011] Yet a further object of the present invention is to provide a vented cap for use on an intermediate bulk container wherein the lower portion has formed therearound a plurality of vent apertures wherein the vent apertures are fluidly coupled to the annular passage.
[0012] To the accomplishment of the above and related objects the present invention may be embodied in the form illustrated in the accompanying drawings. Attention is called to the fact that the drawings are illustrative only. Variations are contemplated as being a part of the present invention, limited only by the scope of the claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0013] A more complete understanding of the present invention may be had by reference to the following Detailed Description and appended claims when taken in conjunction with the accompanying Drawings wherein:
[0014] FIG. **1** is a bottom perspective of the present invention; and
[0015] FIG. **2** is a side view of the present invention; and
[0016] FIG. **3** is a bottom view of the present invention.
DETAILED DESCRIPTION
[0017] Referring now to the drawings submitted herewith, wherein various elements depicted therein are not necessarily drawn to scale and wherein through the views and figures like elements are referenced with identical reference numerals, there is illustrated a vented container cap **100** constructed according to the principles of the present invention.
[0018] An embodiment of the present invention is discussed herein with reference to the figures submitted herewith. Those skilled in the art will understand that the detailed description herein with respect to these figures is for explanatory purposes and that it is contemplated within the scope of the present invention that alternative embodiments are plausible. By way of example but not by way of limitation, those having skill in the art in light of the present teachings of the present invention will recognize a plurality of alternate and suitable approaches dependent upon the needs of the particular application to implement the functionality of any given detail described herein, beyond that of the particular implementation choices in the embodiment described herein. Various modifications and embodiments are within the scope of the present invention.
[0019] It is to be further understood that the present invention is not limited to the particular methodology, materials, uses and applications described herein, as these may vary. Furthermore, it is also to be understood that the terminology used herein is used for the purpose of describing particular embodiments only, and is not intended to limit the scope of the present invention. It must be noted that as used herein and in the claims, the singular forms “a”, “an” and “the” include the plural reference unless the context clearly dictates otherwise. Thus, for example, a reference to “an element” is a reference to one or more elements and includes equivalents thereof known to those skilled in the art. All conjunctions used are to be understood in the most inclusive sense possible. Thus, the word “or” should be understood as having the definition of a logical “or” rather than that of a logical “exclusive or” unless the context clearly necessitates otherwise. Structures described herein are to be understood also to refer to functional equivalents of such structures. Language that may be construed to express approximation should be so understood unless the context clearly dictates otherwise.
[0020] References to “one embodiment”, “an embodiment”, “exemplary embodiments”, and the like may indicate that the embodiment(s) of the invention so described may include a particular feature, structure or characteristic, but not every embodiment necessarily includes the particular feature, structure or characteristic.

[0021] Referring in particular to the Figures submitted herewith, the vented container cap **100** includes a body **10** having an upper portion **12** and a lower portion **15**. The upper portion **12** and lower portion **15** are integrally formed and it should be understood within the scope of the present invention that the body **10** is manufactured from plastic or other suitable material. The upper portion **12** is planar in manner having an upper surface **20** and a lower surface **22**. The upper portion **12** is perpendicular to the lower portion **15** and includes a perimeter edge **25** that extends outward past the wall **30** of the lower portion **15**. The perimeter edge **25** defines the shape of the upper portion **12** wherein in a preferred embodiment of the present invention that the shape of the upper portion **12** is hexagonal. While the upper portion **12** is hexagonal in shape in the Figures submitted herewith, it is contemplated within the scope of the present invention that the upper portion **12** could be provided in alternate shapes. The upper portion **12** provides an element for a user to grasp the vented container cap **100** and either tighten or loosen onto an IBC tote or other container to which the vented container cap **100** is operably coupled.
[0022] The lower portion **15** is contiguous with the lower surface **22** of the upper portion **12** extending outward therefrom being perpendicular thereto. The lower portion **15** is annular in shape having wall **30** wherein the wall has integrally formed therewith the bottom member **40**. The wall **30** includes exterior surface **32** wherein the exterior surface **32** has thread members **34** formed thereon. Thread members **34** facilitate the ability for the vented container cap **100** to be releasably secured to a container opening having mateable threads. A lower aperture **50** is formed in the bottom member **40** and a lower section **37** of the wall **30** distal to the upper portion **12**. Lower aperture **50** is in fluid communication with void **52** wherein void **52** is disposed within the interior of the lower portion **15**. Formed within the wall **30** is annular passage **60**. Annular passage **60** is formed completely within the wall **30** extending fully therearound and is hollow providing fluid communication with the void **52**.
[0023] Annular passage **60** extends substantially the height of the wall **30** and is in fluid communication with vent apertures **70**. Vent apertures **70** are formed within the wall **30** proximate the upper end **33** thereof. The vent apertures **70** are formed around the wall **30** being approximately ninety degrees apart so as to have four vent apertures **70** formed in the wall **30**. The vent apertures **70** being in fluid communication with the annular passage **60** which is in fluid communication with void **52** permits fluid communication with the interior volume of the container to which the vented container cap **100** is secured via lower aperture **50** wherein the lower aperture **50** is disposed within the container and as such fluidly coupled with the interior volume of the container with which the vented container cap **100** is secured.
[0024] The vent apertures **70** permit venting of the container to which the vented container cap **100** is secured and further inhibits any introduction of materials into the container due to the location of the vent apertures **70**. The vent apertures **70** are adjacent the lower surface **22** of the upper portion **12**. Subsequent installation, the vent apertures **70** are in a position such that the upper portion **12** is providing coverage thereof so as to inhibit an ability for debris and materials from entering the vent apertures **70**. While four vent apertures **70** are discussed and illustrated herein, it is contemplated within the scope of the present invention that the vented container cap **100** could have more or less than four vent apertures **70**. The location of the vent apertures **70**, which are in fluid communication with the annular passage **60**, further inhibits liquids disposed in the container to which the vented container cap **100** is secured from splashing out.
[0025] In the preceding detailed description, reference has been made to the accompanying drawings that form a part hereof, and in which are shown by way of illustration specific embodiments in which the invention may be practiced. These embodiments, and certain variants thereof, have been described in sufficient detail to enable those skilled in the art to practice the invention. It is to be understood that other suitable embodiments may be utilized and that logical changes may be made without departing from the spirit or scope of the invention. The description may omit certain information known to those skilled in the art. The preceding detailed description

is, therefore, not intended to be limited to the specific forms set forth herein, but on the contrary, it is intended to cover such alternatives, modifications, and equivalents, as can be reasonably included within the spirit and scope of the appended claims.

Claims

1. A vented container cap configured to be operably coupled to a container and provide venting thereof and further inhibit clogging of the vented container cap from coagulation wherein the vented container cap comprises: a body, said body having an upper portion and a lower portion, said upper portion and said lower portion being integrally formed, said lower portion having an upper end and a lower end, said upper portion being proximate said upper end of said lower portion, said upper portion being perpendicular to said lower portion, said upper portion having an upper surface and a lower surface, said upper portion having a perimeter edge defining a shape thereof, said upper portion having a width greater than a width of said lower portion, said lower portion being cylindrical in shape extending outward from said lower surface of said upper portion, said lower portion having a wall, said wall having a bottom member integrally formed therewith, said bottom member being distal to said upper portion, said wall having an exterior surface, said bottom member having a lower aperture formed therein, wherein said lower aperture further extends onto said wall proximate said lower end of said wall, said lower aperture being in fluid communication with a void present in said lower portion, said lower portion further having an annular passage, said annular passage being within said wall of said lower portion, said annular passage extending completely around said wall, said annular passage being in fluid communication with said void; and a plurality of vent apertures, said plurality of vent apertures being formed through said wall of said lower portion, said plurality of vent apertures being proximate said upper end of said lower portion, said plurality of vent apertures being in fluid communication with said annular passage, said plurality of vent apertures operable to provide venting of the container.
2. The vented container cap configured to be operably coupled to a container and provide venting thereof as recited in claim 1, wherein said exterior surface of said wall further includes thread members formed thereon.
3. The vented container cap configured to be operably coupled to a container and provide venting thereof as recited in claim 2, wherein said upper portion of said body is hexagonal in shape.
4. The vented container cap configured to be operably coupled to a container and provide venting thereof as recited in claim 3, wherein said vent apertures are located around said lower portion having a separation of ninety degrees.
5. The vented container cap configured to be operably coupled to a container and provide venting thereof as recited in claim 4, wherein said vent apertures are located proximate said lower surface of said upper portion at said upper end of said wall of said lower portion.
6. The vented container cap configured to be operably coupled to a container and provide venting thereof as recited in claim 5, wherein said body is manufactured from plastic.